

Addis Ababa Science and Technology University

College of Social and Natural Science

Department of Mathematics

Course Code: Math1007 (Test-1)

Total mark: 10%

Time allowed: 40 minutes

Name Abel Tessema IdNo: 0027/14 Section A1

I. (Short answer): write your answer in the space provided (each 1pt)

1. $\neg(p \wedge q) \equiv \neg p \wedge \neg q$ (T/F) F
2. For any propositions p and q , $(p \Rightarrow q) \Leftrightarrow (p \wedge \neg q)$ is tautology (T/F) F contradiction
3. Let $U = z$

Let $P(x, y) = 2x + y = 10$, then, what is the truth values of

- a) $(\forall x)(\forall y)P(x, y)$ F
 - b) $(\forall x)(\exists y)P(x, y)$ T
 - c) $(\exists x)(\forall y)P(x, y)$ F
 - d) $(\exists x)(\exists y)P(x, y)$ T
4. Let $A = \{a, b, c, d, e\}$ and $B = \{1, 2, 3, d, e\}$ then, what are the elements of $A \Delta B = \{a, b, c, 1, 2, 3\}$
 5. Let $A = \{a, \{b, c\}, d\}$, then, $P(A) = \{\{a\}, \{\{b, c\}\}, \{d\}, \{a, \{b, c\}\}, \{\{b, c\}, d\}, \{a, d\}, \{a, \{b, c\}, d\}, \emptyset\}$

II. Workout: Try to show the necessary steps clearly

1. Show the validity of the following arguments

(2pt)

If it is raining then streets are wet

It is raining

Therefore, streets are wet

Let p : It is raining

q : The streets are wet.

Then the argument can be written as:

$$p \Rightarrow q, p \vdash q$$

Let's check the validity:

1) $p \Rightarrow q$ is true --- Given

2) p is true --- Given

3) q is true --- From ① & ②, by the rule of implication.

Therefore, the argument is valid.

OR

Let's try to construct a truth table.

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Only on the first row are both ' p ' and ' $p \Rightarrow q$ ' true. And on that row, ' q ' is also true. Since it's impossible for the premises to be true & the conclusion false, the argument is valid.